CampusGpt

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Abstract:

Smart campuses demand faster and more efficient academic and administrative support to address the increasing demands of the student and academic communities. Conventional manual approaches are often slow, reactive, and unable to manage high volumes of queries efficiently. In this context, this research proposes the use of CampusGPT, which is a fully autonomous AI campus assistant that utilizes the power of a multi-agent artificial intelligence paradigm.

CampusGPT is a novel initiative that utilizes the power of specialized AI agents that facilitate the accomplishment of important tasks like tae generation, automated question paper generation, career guidance, and intelligent question solving. By employing the power of natural language processing and machine learning technologies, the proposed solution is able to offer personalized and context-aware support, thereby reducing the workload and improving the efficiency of the campus.

Keywords: CampusGPT, Smart Campus, Autonomous AI Assistant, Multi-Agent System, Artificial Intelligence, Natural Language Processing

Introduction:

The accelerated process of digital transformation for higher education institutions has led to the development of smart campuses that are intended to be intelligent, efficient, and student-centric. However, existing campus infrastructure for providing such services to students has been found to be mostly manual, disjointed, and reactive. This has caused delays in providing academic assistance, decision-making for administration, and providing personalized guidance to students. Students and faculty need timely assistance with matters such as examination preparation, assignment development, career guidance, and other academic-related matters. To this effect, this research seeks to develop a smart campus assistant service dubbed CampusGPT, which will be an autonomous AI assistant based on a multi-agent system.

The proposed system, CampusGPT, is a modular intelligent system, and multiple agents operate collaboratively using different machine learning and pre-trained language models. There is a question paper generation agent using AI and a pre-trained instruction model to generate structured question papers from the syllabus, which is stored as a JSON file. There is also an assignment generation agent using a combination of semantic topic extraction and a light-weight local language model to generate difficulty-aware questions from the notes of the faculty. Further, there is a career guidance agent using a supervised machine learning model and a Random Forest Classifier to generate personalized career paths and identify skill gaps using resume parsing and feature engineering.

The retrieval-augmented conversational agent extends the functionality of intelligent query resolution by incorporating embedding-based semantic search with customizable large language model providers. FastAPI is used for developing backend services that support scalable API communication for agents, along with a cross-platform interface based on Flutter for smooth user interaction. Through the integration of various AI disciplines such as NLP, semantic retrieval, and predictive analytics, CampusGPT hopes to minimize administrative burden, improve academic productivity, and help achieve the vision of intelligent smart campus ecosystems.

Methodology:

The system, termed CampusGPT, is based on a multi-agent artificial intelligence system that enables the automation of academic assistance in a smart campus setting. The methodology is based on a four-fold framework that includes various functional modules, as represented by workflow diagrams. These modules comprise the career guidance agent, assignment generation agent, intelligent query resolution agent, and question paper generation agent. The system, in essence, follows various stages that involve data acquisition, preprocessing, intelligent inference, and result delivery, as enabled by FastAPI and Flutter.

1.Career Guidance Agent:

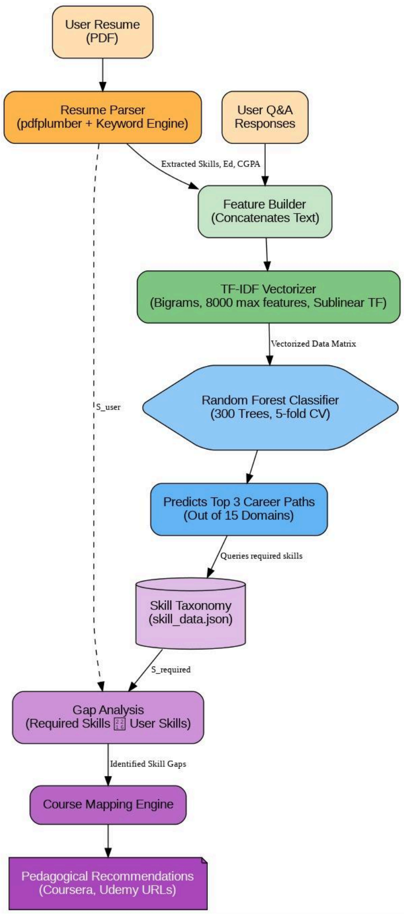

Fig.1. Career Guidance Agent Workflow

The workflow for the career guidance process, as represented in Fig-1, starts with the collection of student resumes in PDF format along with the questionnaire responses provided by the users. The system uses various document parsing techniques to identify the key attributes such as skills, educational background, and academic performance. These attributes are then integrated with the help of a feature construction module to form a comprehensive profile for the students. The textual profile for the students is then converted into a numerical format with the help of a TF-IDF vectorizer along with a contextual n-gram representation. The numerical features are then passed to a supervised ensemble learning model to identify the best possible career options from a set of predefined domains. The career domain identified by the model for the student is then compared with a structured skill taxonomy repository to identify the skill gaps. The skill gaps are then used to identify the learning resources for the students.

2. Assignment Generation Agent:

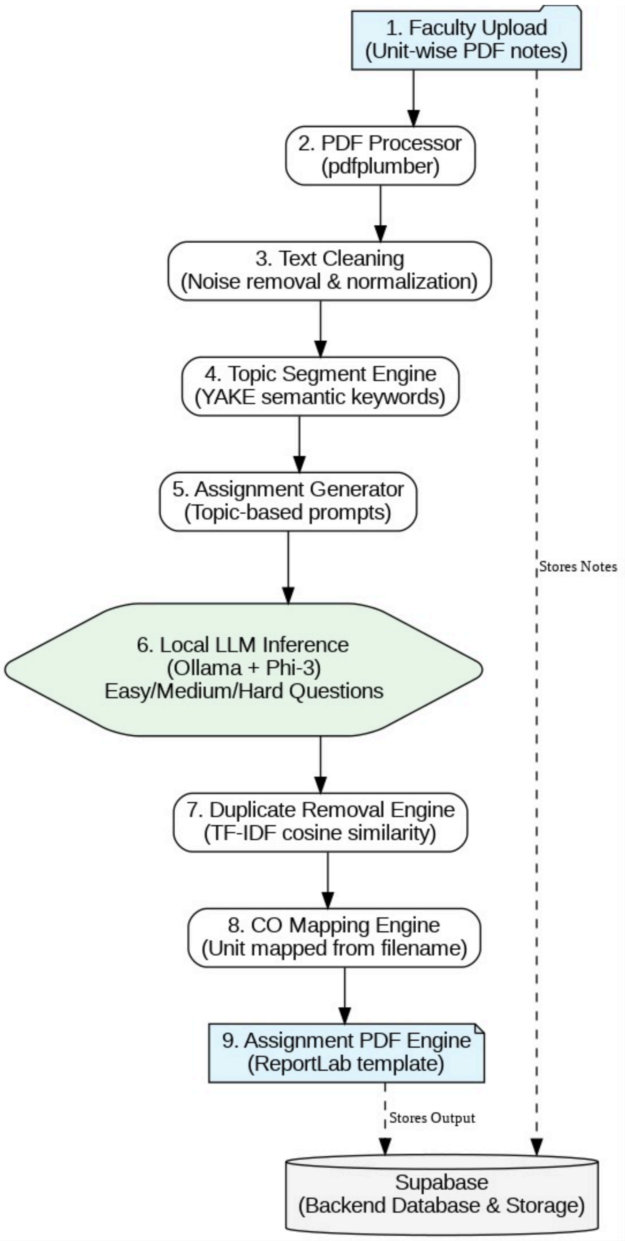

Fig.2. Assignment Generation Agent

Fig-2 illustrates the workflow for automated assignment generation. Within this module, instructors are required to upload academic notes in unit-wise PDF format. The system will then identify content from the uploaded files and undertake necessary preprocessing tasks such as removing noise from the content. A semantic topic segmentation mechanism will then be used to identify relevant academic topics from the uploaded content. The identified topics will then be used to generate optimized prompts that will be processed using a local language model to generate assignment questions of varying difficulty levels. To ensure diversity in content, similarity-based filtering mechanisms will be used to filter duplicate content. The generated content will then be mapped to relevant course outcomes in a structured assignment document.

3.Question Paper Generation Agent:

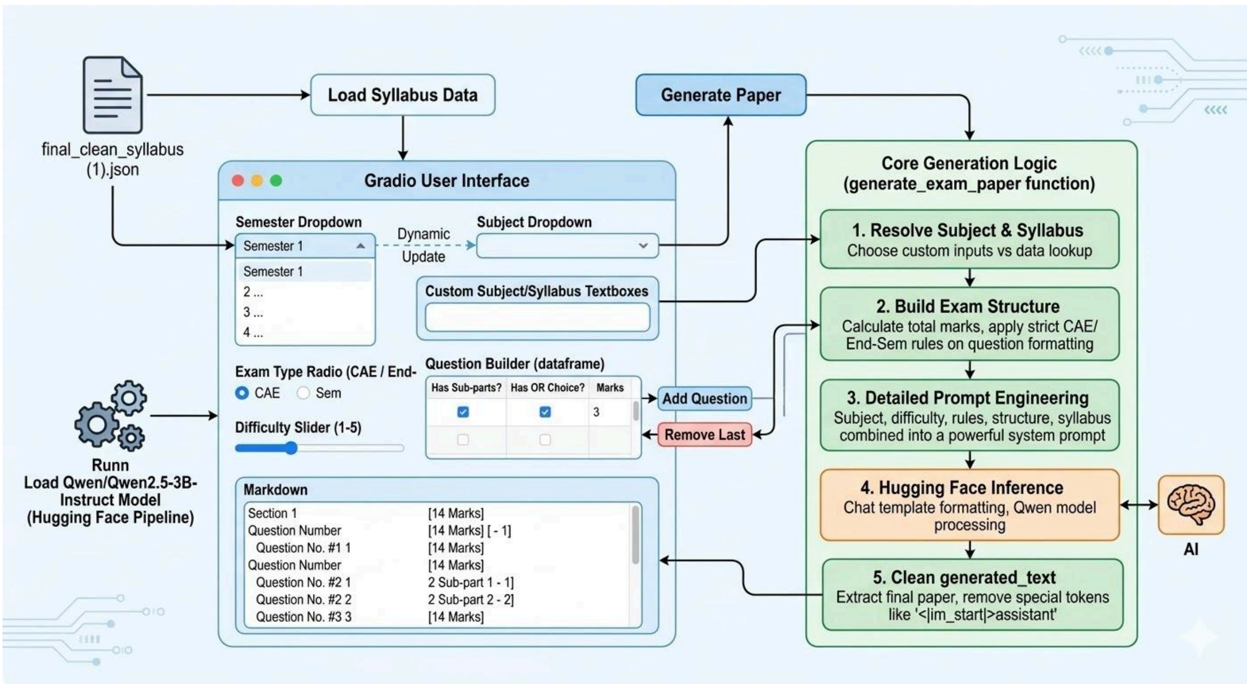

Fig.3.Question Paper Generation Agent

Fig.3 depicts the workflow of the automated question paper generation module. The system begins by loading semester-wise data on the syllabus, which is stored in a structured JSON file. The system then takes input parameters such as subject selection, examination type, difficulty level, and question structure requirements. The system then uses these parameters to generate the examination blueprint in an automated way by calculating the distribution of marks and applying formatting requirements such as sub-parts and optional questions. The system then generates a prompt that includes the content of the syllabus and the structure requirements, and then uses an instruction-tuned language model to generate the examination questions. The generated question is then post-processed to remove unwanted tokens and then formatted in a standard question paper style.

4.Intelligent Query Resolution Agent:

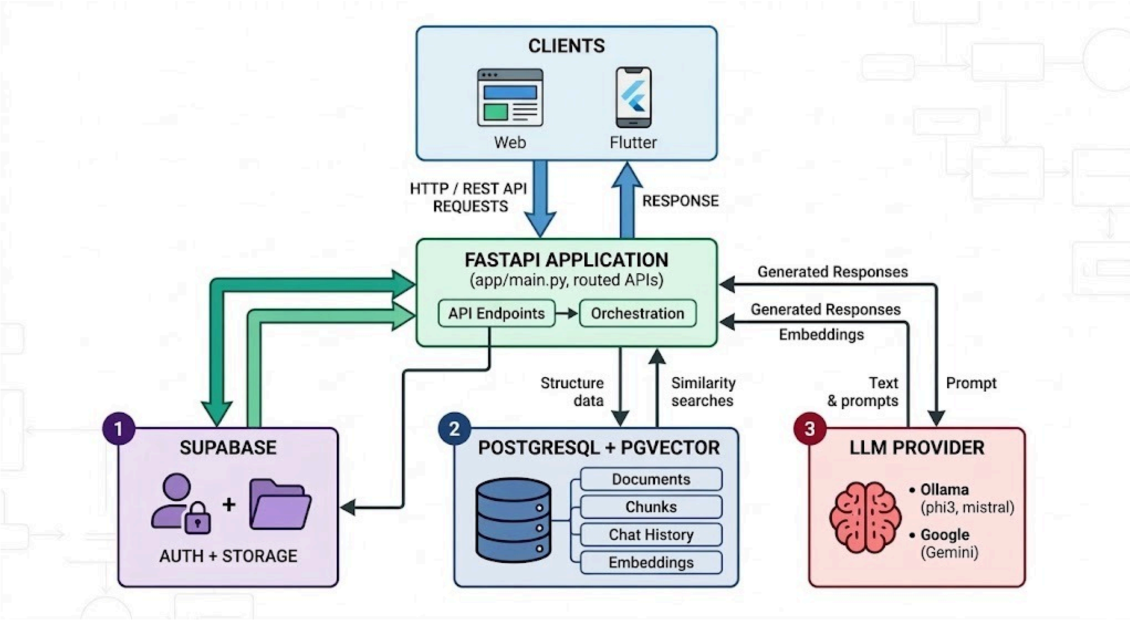

Fig.4. Intelligent Query Solver Agent

The intelligent query resolution process illustrated in Figure 4 enables real-time academic assistance through conversational interaction. In this process, the institutional documents and academic resources are uploaded and processed into smaller segments of text. These segments are then converted into vector representations using semantic representation models. They are then stored in a vector-enabled database. When a user sends a query through the application interface, the query is converted into a vector representation and used to retrieve the most relevant contextual information. The knowledge is then used to create a prompt that is processed by a large language model to create context-aware responses.

Output and Results:

Discussion:

The proposed system, CampusGPT, demonstrates the possibilities of developing an effective multi-agent framework based on an artificial intelligence system that can be used to automate key academic support services in smart campuses. The system, based on machine learning, language models, and semantic retrieval, aims to offer effective services in the form of personalized career counseling, automated assignment and question paper generation, and query answering.

The proposed system is still in the development stage and needs to be improved and integrated in the future to evaluate its performance and efficiency in meeting the requirements of an effective smart campus system. The proposed system, once improved, can be effective in developing an effective system that can be used to move towards an intelligent, automated, and student-centric campus system.

Conclusion:

This research presents CampusGPT, a multi-agent artificial intelligence framework designed to automate academic and administrative support services in smart campus environments. The proposed system integrates different intelligent modules for career guidance, assignment generation, question paper creation, and conversational query resolution. By combining machine learning techniques, large language models, and semantic retrieval mechanisms, the system aims to improve efficiency, reduce manual workload, and provide personalized academic assistance to students and faculty.

Although the system is currently in the development stage, the proposed architecture demonstrates the feasibility of building an intelligent and scalable campus support platform. With further integration, dataset enhancement, and model optimization, CampusGPT has the potential to transform traditional campus operations into proactive, data-driven, and user-centric digital services, contributing to the realization of fully intelligent smart campus ecosystems.

Future Scope:

The proposed CampusGPT system may be further enhanced with the aim of improving the performance of individual AI agents that will ultimately lead to the development of a fully autonomous and real-time academic support platform. Future enhancements may be directed towards fine-tuning the performance of the language models with respect to the quality and relevance of the generated content, including assignments and examination papers. Furthermore, the proposed CampusGPT system may be enhanced with the aim of improving the performance of individual AI agents that will ultimately lead to the development of a fully autonomous and real-time academic support platform.

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